

Shift Turnover Training — Research Report & Implementation Strategy

MedSim V8 / VRAI Faces · Functional Requirement **FR-009** · Prepared for instructor review

Date: 2026-06-12 · **Status:** DRAFT — awaiting instructor ratification · **Companion plan:** docs/PLAN-2026-06-12-fr009-shift-handoff.md · **Research basis:** 22 sources (AHRQ, PubMed Central, Wiley JAN, AHRQ TeamSTEPPS), 109 extracted claims, 5 search angles

Builds on shipped machinery: chart engine (MAR, allergies, vitals trends, notes, care team) · staged-error system (FR-008, incl. the "report" encounter) · room-local speech transcription (FR-006b) · AI character prompting (FR-001/002/003) · comparison/rubric store · debrief builder · two-stage control room (FR-005)

1 · The requirement (instructor, 2026-06-12)

Critical training at the start and end of the shift: shift turnover. For single-patient or multiple-patient sessions there needs to be an end-of-shift handoff. This can be a scenario itself, built using the other scenarios. The existing portfolio — and the future expanded portfolio — provides the context: current patient state, concerns, key medications and treatments, things to look out for. Used to check what follow-up questions a student generates if oncoming, or how they respond if off-going. After the handoff, a student survey the student answers with recorded verbal responses, evaluated on their perceptions versus what detail and questions actually came up, key gaps, and whether any high-risk elements were missed that would put a patient at risk. The oncoming/off-going nurse or charge nurse comes from the existing character list.

2 · Executive summary

- **Recommended framework:** an SBAR-skeleton element checklist (nursing-native, what students meet in practice) **enriched with the two elements the evidence says matter most and students miss most:** an up-front illness-severity statement and a closing receiver *synthesis* (read-back) — plus anticipatory guidance ("if X happens, do Y") as a first-class, high-risk element, and an explicit responsibility-transfer moment.
- **Recommended scoring:** binary element-coverage of the handoff transcript against a per-patient ground-truth "context pack" generated from the live chart — the design lesson from validated instruments is that binary coverage scores reliably while "partial credit" does not. AI-assisted scoring, instructor-confirmed, formative not summative.
- **The survey closes the perception loop:** evidence shows self/peer assessment runs systematically high versus external evaluation; the post-handoff verbal survey (answered by voice on the station, transcribed by the room transcriber) captures the student's *perception*, which the evaluator then sets against the *measured* coverage map — gaps and missed high-risk elements named explicitly.
- **Direct synergy with the staged-error system:** the handoff IS FR-008's "report" encounter point — an armed discrepancy can ride the handoff, and the survey probes whether the student noticed.

- **Effort:** six stages (H1–H6), all server/portal-side plus one station flow; no new hardware, no new privacy surface (survey audio rides the already-ratified room-local transcription path).

3 • Research findings

Verification status — read first. The research harness fetched 22 sources and extracted 109 claims, but its adversarial cross-verification stage hit a usage cap and could not complete; the findings below are therefore **cited extractions from primary sources, not formally cross-verified**. They are internally consistent across independent sources and consistent with the preparer's domain knowledge. Items load-bearing for the design are marked **KEY** and recommended for instructor spot-check via the cited link.

3.1 Framework landscape (what structure to teach)

- **KEY** The AHRQ *Making Healthcare Safer IV* review of structured handoffs rates **I-PASS** with the strongest certainty of patient-safety benefit (moderate; including RCT evidence) and **SBAR** lower (low certainty; a third of studies showing no benefit). [1][2]
- **KEY** The same literature is **profession-split**: SBAR evidence comes predominantly from nursing handoffs; I-PASS evidence predominantly from physician signout — neither generalizes automatically to the other. For a NURSING shift-report trainer this argues for an SBAR-shaped skeleton carrying I-PASS's strongest elements, not a wholesale I-PASS adoption. [2]
- I-PASS benefits were demonstrated as a **bundle** — mnemonic + workshop + simulation practice + observation tools — not the mnemonic alone. The platform's contribution is exactly the expensive part of that bundle: repeatable simulated practice with observation. [1]
- Nursing-specific I-PASS adaptations are emerging (e.g., a 2025 *Journal of Advanced Nursing* pilot structuring oral end-of-shift bedside handover, reporting improved handover quality across ~800 evaluated handovers). The space is converging on hybrid structures — which is what we recommend. [3]
- TeamSTEPPS defines a handoff as transferring **information + authority + responsibility** — so the checklist ends with an explicit acceptance moment, not just a data dump. [4]

3.2 Validated assessment instruments (how to score)

Instrument	Shape	What we take from it
Handoff CEX (Yale; validated on nursing shift report)	6 domains (setting, organization, communication, content, judgment, professionalism) + overall; 9-point scale banded unsatisfactory/satisfactory/superior; paired provider and recipient versions	The domain list for instructor-facing summary; the precedent that the RECEIVER is assessed too — but its receiver version performed weakest (narrow range, low agreement), so receiver assessment needs its own design (question quality + synthesis), which our oncoming mode provides. [5][6]

SBAR-LA (learner assessment)	10 items scored binary 0/1 ; developers abandoned a 0–2 partial-credit scale because raters could not reliably distinguish partial from full	The central automation lesson: binary element coverage is the reliable scorable unit — exactly what a transcript comparator can do well. [7]
SBAR nursing change-of-shift instrument (content-validated, CVI $\approx$ 92%)	Element checklist: vitals, oxygenation, level of consciousness, mobility, drains/catheters, exams, nutrition, dressings, eliminations, medications, complications; allergies under Background	Seed list for our context-pack elements (merged with I-PASS severity + contingencies). [8]

3.3 What untrained students actually miss (failure modes to target)

- **KEY** In 177 simulated signouts, **anticipatory guidance / contingency planning was omitted in ~54%** (up to 84% for some cases) — the single dominant high-risk omission. [9]
- **KEY** **Receiver synthesis almost never happens spontaneously** (1 of 30 participants, once) — receivers ask questions but do not read back. Training must force the synthesis moment; our AI counterpart explicitly asks for it. [9]
- Untrained completeness ran $\approx$ 52% against an expert element checklist, and completeness correlated strongly with overall quality ($R^2 \approx 0.9$) — **supporting checklist-coverage as the automated scoring proxy**. [9]

3.4 The perception gap (why the survey works)

- Peer/insider ratings of handoff quality run systematically **higher than external evaluators'** — the published basis for expecting students to overestimate their own completeness. The survey is built to surface exactly that delta: it asks the student to *predict* their coverage, then the evaluator compares prediction to the measured map. [6]

3.5 High-risk element list (synthesis of the instruments above)

Omissions flagged **HIGH-RISK** in the evaluator (the instructor's "would put a patient at risk" list): **allergies** · **code status** · **medication changes** (new/held/dose-changed/last-given/due-next, infusions running) · **pending results or tasks** · **lines/drains/access** · **active safety risks** (falls, airway, sepsis watch, bleeding precautions) · **anticipatory guidance / contingencies** · **abnormal vitals trend** · any **active staged discrepancy** (FR-008).

4 · The platform design

4.1 The handoff context pack — the ground truth

Generated per patient from the live session (chart seed + events + medication board + staged-error state) at handoff time. It is simultaneously: the AI counterpart's knowledge, the completeness rubric, and the debrief artifact.

#	Element	Source in the system	Risk
1	Identity & situation (name, age, room, admitting problem, today's story)	chart banner, chief complaint, encounter	
2	Illness severity, one line (stable / watcher / unstable)	vitals trend vs condition ranges	HIGH
3	Background: diagnoses, relevant history, allergies , code status	problem list, allergies, code_status	HIGH
4	Current assessment: latest vitals + trend direction, key labs, focused exam notes	vitals_baseline + vitals.record events, labs	
5	Medications & treatments: new / changed / held / due-soon, infusions running, high-alert flags	MAR + admin history, med board, IV fluids	HIGH
6	Lines / drains / access	chart (iv_fluids, devices)	HIGH
7	Pending items: orders, labs, consults awaiting	orders, events	HIGH
8	Safety risks (falls, airway, bleeding, isolation...)	condition + overlays + safety_class	HIGH
9	Anticipatory guidance: "watch for X — if X, then Y"	condition profile + active staged errors	HIGH
10	Receiver synthesis (read-back) occurred	transcript	HIGH
11	Responsibility transfer explicit ("I've got them")	transcript	

4.2 Two student roles, one machinery

Student OFF-GOING (gives report)		Student ONCOMING (receives report)
AI counterpart	Oncoming nurse / charge nurse (from the existing character list) — holds the context pack, listens, then probes 2–4 omissions prioritizing high-risk ones, and asks for a synthesis if the student doesn't offer one	Off-going nurse — delivers report at an instructor-set completeness dial : complete · typical-gaps (the evidence-backed common omissions: anticipatory guidance + a pending item) · staged-error embedded (FR-008 report encounter)
Assessed behavior	Element coverage of what the student SAID + response quality to probes	The QUESTIONS the student asks (mapped to the gap list), read-back/synthesis, and whether planted gaps/discrepancies were caught

Containment	Same rule as the staged-error system: the AI introduces nothing beyond its instructions, never hints anything is staged, stays in character
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4.3 Multi-patient mode (charge-nurse turnover)

Sequential context packs for 2–3 patients (v1 cap, ratify) + one cross-patient element: **prioritization** — "who do you see first and why." Builds on the shared-staff investigation (FR-007): one charge-nurse character handles all patients' reports in one conversation.

4.4 The post-handoff verbal survey

Presented on the station immediately after the handoff; the student answers each question by holding to talk; the room transcriber (already ratified, in-room only) writes the answers down. Questions (v1 set, ratify):

1. "Overall, how complete was the handoff you just gave/received, 0–10 — and why?" (*prediction*)
2. "Name the three most important things you handed off / heard." (*prioritization recall*)
3. "What, if anything, do you think you missed or wish you'd asked?" (*self-identified gaps*)
4. "What should be watched for on this patient over the next shift?" (*anticipatory guidance recall — the #1 evidence-backed omission*)
5. "Was there anything in the chart or report that didn't match what you heard or read?" (*discrepancy probe — fires the FR-008 comparison when an error was staged*)
6. *Oncoming only*: "What is your first action for this patient?" (*synthesis into plan*)

4.5 Evaluation & debrief

- **Coverage map**: each pack element scored binary (said / not said) against the handoff transcript — AI-assisted on the existing comparison-store pattern, every line instructor-confirmable before it is shown to the student. High-risk misses called out in red.
- **Perception delta**: the survey's self-rating and claimed coverage set against the measured map — "you rated yourself 8/10 and named meds and vitals; the record shows allergies, the pending culture, and the watch-for items were never mentioned."
- **Receiver metrics (oncoming)**: questions asked vs the gap list; synthesis performed y/n; planted gap/discrepancy caught y/n.
- **Debrief artifact**: one section in the existing debrief surface — pack, transcript, coverage map, survey answers, perception delta, high-risk misses — printable for post-sim discussion. Formative, not a grade (recommended).

4.6 Composition (the instructor's "scenario built from scenarios")

Two entry points, same engine: **(a)** a handoff PHASE appended to any live session ("end the shift" button in the control room); **(b)** a STANDALONE handoff session built from the charts of one or more existing scenarios — pick patients on the Setup stage, the engine seeds their charts, the session IS the turnover.

5 • Implementation strategy

Stage	Scope	Builds on (already shipped)	Effort
H1	Context-pack generator + high-risk tagging + ground-truth element list	chart seed/fold, med board, FR-008 state	M
H2	Handoff session mode + AI counterpart prompts (receiver-probe / giver-with-dial, containment, synthesis ask, responsibility transfer)	FR-001/002/003 prompt machinery, persona list	M
H3	Multi-patient sequencing + prioritization element	H1–H2, FR-007 direction	M
H4	Verbal survey station flow (question sequence → PTT answers → storage)	FR-006b room transcription, audio station	M
H5	Evaluation engine (coverage map, survey comparison, perception delta, instructor confirm) + debrief section	comparison store, debrief builder	L
H6	Control-room integration (Setup: phase/role/dial/characters · Live: start-handoff + status) + field validation script	FR-005 two-stage control room	M

Sequencing: FR-008's remaining stages (S5 builder page, S6 debrief) come first — H5/H6 share those surfaces. Then H1→H6 in order; H1+H2 alone already yield a usable single-patient handoff drill.

6 · Risks & open questions

- **Scoring validity:** binary coverage measures completeness, not judgment quality — mitigated by instructor confirmation and formative use. *Ratify: formative-only for v1.*
- **AI receiver question quality:** the probing behavior needs prompt iteration in field tests (same path FR-001's decisive-doctor took).
- **Survey transcription accuracy:** drug names in answers ride the existing vocabulary-hints lever; the pack's terms join the hints during survey mode.
- **Session length (multi-patient):** cap at 2–3 patients for v1. *Ratify.*
- **Verification caveat** on the research basis (Section 3 box): key claims flagged for spot-check.

7 · References

- [1] AHRQ Making Healthcare Safer IV — Structured Handoffs rapid review. effectivehealthcare.ahrq.gov/sites/default/files/related_files/structured-handoff-rapid-research.pdf
- [2] BMJ Quality & Safety (2025) — MHS IV structured-handoff certainty ratings. pmc.ncbi.nlm.nih.gov/articles/PMC12232517/
- [3] Journal of Advanced Nursing (2025) — I-PASS-structured bedside nursing handover pilot. onlinelibrary.wiley.com/doi/10.1111/jan.16896
- [4] AHRQ TeamSTEPPS 3.0 — Handoff (information + authority + responsibility). ahrq.gov/teamstepps-program/curriculum/communication/tools/handoff.html
- [5] Handoff CEX — nursing shift-report validation. pmc.ncbi.nlm.nih.gov/articles/PMC3504166/
- [6] Handoff CEX — multi-site validation, rater effects, receiver-version limits. pmc.ncbi.nlm.nih.gov/articles/PMC3621018/
- [7] SBAR-LA — 10-item binary learner-assessment rubric. ncbi.nlm.nih.gov/pmc/articles/PMC8520891/
- [8] SBAR change-of-shift instrument — content validation + element checklist. ncbi.nlm.nih.gov/pmc/articles/PMC9749775/

[9] Simulated signout analysis (2024, n=177) — contingency omissions, receiver synthesis, completeness↔quality.
ncbi.nlm.nih.gov/pmc/articles/PMC11430516/

Additional consulted sources (13 more, incl. TeamSTEPPS curriculum pages, simulation-pedagogy and multi-patient handoff literature) are listed in the research run log.